Ch.7 Establishing Table Structures

[Step 3: Create data structure]

- Tables == subjects
- Fields == characteristics

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Part A) Three steps to create preliminary table list:

1. Use preliminary field list (why? helps find missing subj.) (from step 2) ➔ ver. 1
2. Use subject list (from step 2) ➔ ver. 2
3. Use mission objectives (from step 1) ➔ ver. 3

1. Use preliminary field list (to create table list)

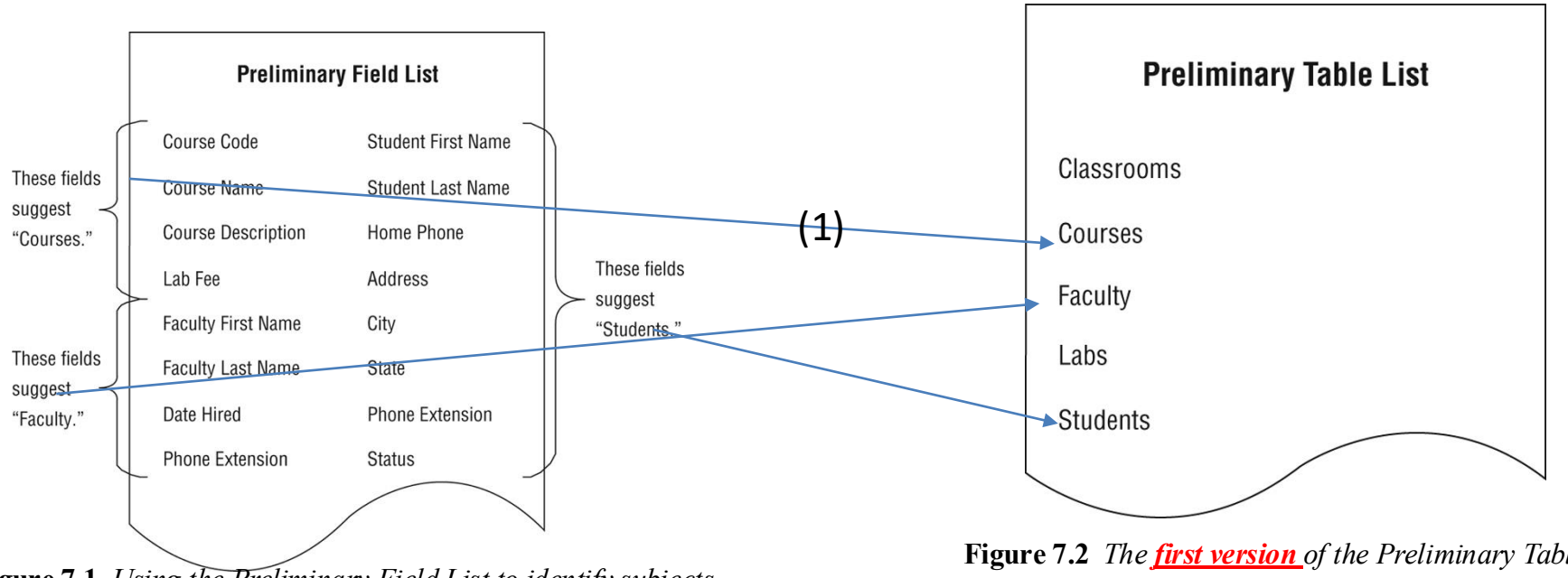

Figure 7.1 *Using the Preliminary Field List to identify subjects*

Figure 7.2 *The first version of the Preliminary Table List*

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2. Use subject list (to create table list)

1. Resolve duplicates that are different - ex. Equipment in both subj. and table list
2. Resolve duplicates that are same – ex. Clients and Customers are the same
3. Combine subj. and table list (copy what is left in subj. list)

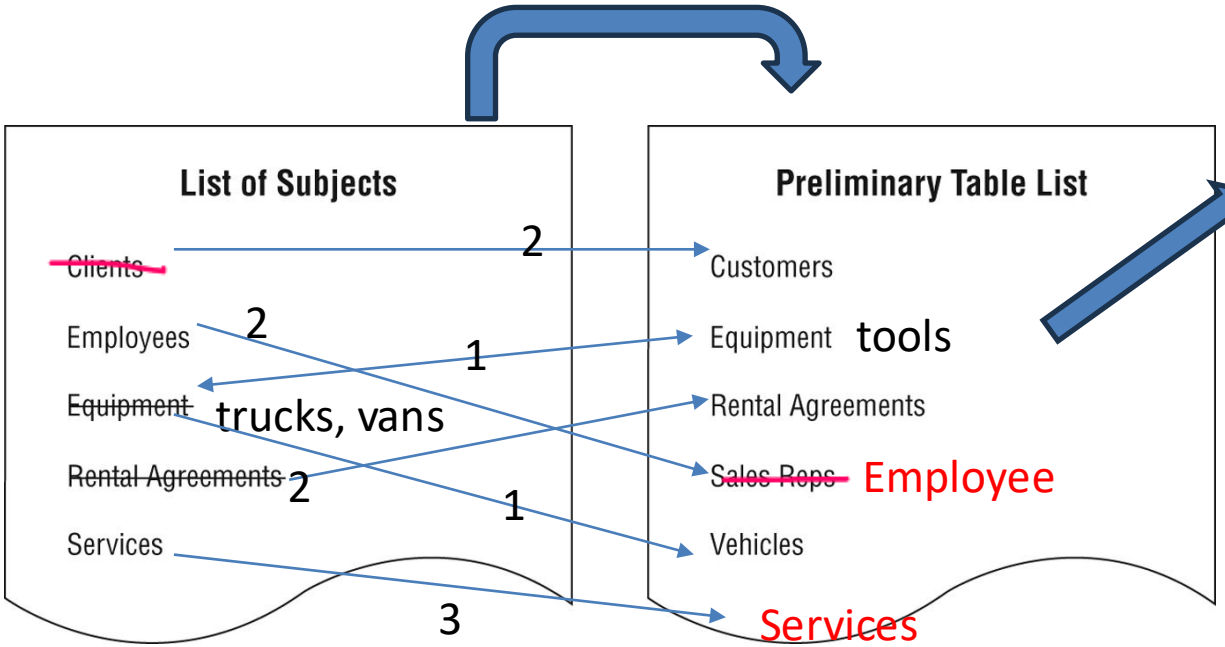

Figure 7.4 *The revised list of subjects and the revised Preliminary Table List (first view)*

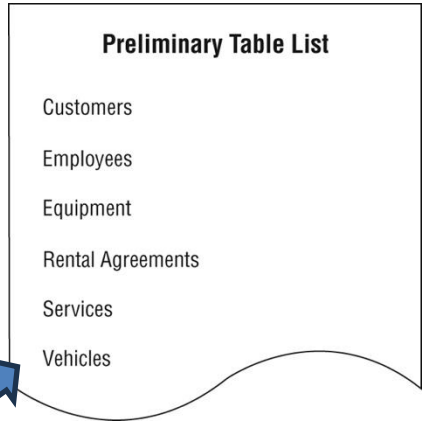

Figure 7.6 *The* ***second version*** *of the Preliminary Table List*

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3. Use mission obj. (to create table list)

1. Have you overlooked any subject?
2. Apply SIT to each mission objective and create an entire new subject list
3. Apply the step 2 from previous slide to merge this new subject list into the preliminary table list – see flight training school ex. in the book

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Part B) Create name, type, description for tables

- 4 table **types**: data, linking, subset, validation
- Table **naming**
  1. Unique & descriptive
  2. Unambiguous – ex. “Dates” is bad; “Entertainer Schedule” is good
  3. Short but full word, no acronyms – ex. Equipment, not EQ
  4. No confusing words – ex. no “Patient Record”, instead three tables, “Patients”, “Doctors”, “Examinations”
  5. No proper names – ex. no “South Region Employee”
  6. No name with more than one subject – ex. no “Department or Branch”
  7. Use **plurals for table** names (and singular for field names) – ex. “Customers”, “Vehicles”

Final Table List

Name	Type	Description
Classrooms	Data	The spaces or areas within a facility reserved for the purpose of conducting class proceedings. Information regarding the physical aspects, on-site resources, and availability of these areas is useful because it allows us to assign classes to the facility that can make the best use of these areas.
Courses	Data	The programs of instruction conducted through courses offered by this institution. Course information must always reflect the addtion of new courses, the deletion of old courses, and the continuing evolution of existing courses.

Figure 7.9 *An example of a Final Table List*

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Table descriptions

- Accurate

Suppliers: the companies that supply us with ingredients and equipment.

Suppliers: the people and organizations from which we purchase ingredients and equipment

- One sentence to explain why the table is needed

...to keep track of the names, addresses, phone numbers, and contact names of all our suppliers.

...it allows us to maintain a constant supply of ingredients and ensure that our equipment is always in working order.

- Clear and succinct

Student Schedule: Those classes that the student is scheduled to attend during this school year. The information provided by this table helps the student implement effective time management and enables the school to figure class loads and student loads.

- No need to say how it is used/implemented

Student Schedule: Those classes that the student is scheduled to attend during this school year. This information is used by the registrar and is accessed from the Student Admissions menu in the Registration Program.

- Don't mention other tables

Dependents: the spouse, children, or wards of a given employee. (See description of Employee table for further information).

When having all table descriptions, interview representatives of users and managers to get their consensus on it.

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Part C) Associate fields with each table

- Use preliminary field list (from step 2) → determine what fields best describe a table.

Table Structures

Courses	Subjects	Instructors	Students

Figure 7.10 *Setting up a sheet for listing table structures*

Table Structures

Classes	Subjects	Instructors	Students
Class Number	Subject Name	Instructor Name	Student Name
Class Name	Subject Description	Instructor SSN	Student SSN
Subject Name	Category	Instructor Address	Student Address
Instructor Name	Credits	Instructor Phone	Student Phone
Room Number		Date Hired	
		Pay Rate	

Figure 7.11 *Listing tables with their associated fields*

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Improve field names

1. Unique & descriptive
2. Unambiguous
3. Short (minimum number of words)
4. No acronyms but ok for few good abbreviations
5. No confusing words
6. No names containing many characteristics – ex. no “Phone/Fax”
7. Use **singulars for field** names (and plurals for table names)

Figure 7.12 *Revised field names*

Table Structures

Classes	Subjects	Instructors	Students
Class Number	Subject Name	Instructor Name	Student Name
Class Name	Subject Description	Instructor SSN	Student SSN
Subject Name	Category	Instructor Address	Student Address
Instructor Name	Credits	Instructor Phone	Student Phone
Room Number		Date Hired	
		Pay Rate	

Table Structures

Classes	Subjects	Instructors	Students
ClsNumber	SubjName	InstName	StdName
ClsName	SubjDescription	InstSocial Security Number	StdSocial Security Number
SubjName	Category	InstAddress	StdAddress
InstName	Credits	InstPhone	StdPhone
Room Number		Date Hired	
		Pay Rate	

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Part D) Elements of the ideal field

1. Represents distinct characteristic of the subject
2. Contains only a single value – ex. not multivalued “phone/fax”
3. Cannot be deconstructed into smaller parts – ex. not multipart “violin, piano”
4. Is not a calculated field
5. Distinct in the entire database
6. Retains its properties when appears in other tables

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Q: identify problematic fields

Table Structures

Instructors

InstName

InstAddress

InstPhone

Categories Taught

InstSocial Security Number

Date Hired

Pay Rate

Figure 7.13 *A table containing fields with questionable structures*

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Q: identify problematic fields

multivalued

Instructors

InstName	InstAddress	InstPhone	Categories Taught	<< other fields >>
Kira Bently	3131 Mockingbird Lane, Seattle, WA 98157	363-9948	DTP, SS, WP	.....
Timothy Ennis	7402 Kingman Drive, Redmond, WA 98115	527-4992	WP, DB, OS	.....
Shannon Black	4141 Lake City Way, Seattle, WA 98136	336-5992	DB, SS	.....
Estela Rosales	970 Phoenix Avenue, Bellevue, WA 98046	322-6992	DTP, WP, PG	.....

multipart

Figure 7.14 *Testing a table with sample data*

Table Structures

Instructors

InstName
InstAddress
InstPhone
Categories Taught
InstSocial Security Number
Date Hired
Pay Rate

Figure 7.13 *A table containing fields with questionable structures*

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Resolve multipart fields = create separate fields

Figure 7.15
Resolving the multipart fields in the INSTRUCTORS table

Instructors

InstFirst Name	InstLast Name	InstStreet Address	InstCity	InstState	InstZipcode	InstPhone	Categories Taught	*<< other fields >>*
Kira	Bently	3131 Mockingbird Lane	Seattle	WA	98157	363-9948	DTP, SS, WP	.....
Timothy	Ennis	7402 Kingman Drive	Redmond	WA	98115	527-4992	WP, DB, OS	.....
Shannon	Black	4141 Lake City Way	Seattle	WA	98136	336-5992	DB, SS	.....
Estela	Rosales	970 Phoenix Avenue	Bellevue	WA	98046	322-6992	DTP, WP, PG	.....

Instructors

InstName	InstAddress	InstPhone	Categories Taught	*<< other fields >>*
Kira Bently	3131 Mockingbird Lane, Seattle, WA 98157	363-9948	DTP, SS, WP	.....
Timothy Ennis	7402 Kingman Drive, Redmond, WA 98115	527-4992	WP, DB, OS	.....
Shannon Black	4141 Lake City Way, Seattle, WA 98136	336-5992	DB, SS	.....
Estela Rosales	970 Phoenix Avenue, Bellevue, WA 98046	322-6992	DTP, WP, PG	.....

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Example of hidden multipart fields

GUIT = Category ("Guitar")
2201 = Identification Number

Instruments

Instrument ID	Manufacturer	Instrument Description	<< other fields >>
GUIT2201	Fender	Stratocaster	.....
MFX3349	Zoom	Player 2100 Multieffects	.....
AMP1001	Marshall	JCM 2000 Tube Super Lead	.....
AMP5590	Crate	VC60 Pro Tube Amp	.....
SFX2227	Dunlop	Cry Baby Wah-Wah	.....
AMP2766	Fender	Twin Reverb Reissue	.....

Figure 7.16 *An example of a "hidden" multipart field*

- Be alert
- How do you retrieve all Guitars manufacturers?

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Resolve multivalued fields (sol. 1)

Instructors

InstFirst Name	InstLast Name	InstStreet Address	InstCity	<< other fields >>	Category Taught 1	Category Taught 2	Category Taught 3
Kira	Bently	3131 Mockingbird Lane	Seattle	.....	DTP	SS	WP
Timothy	Ennis	7402 Kingman Drive	Redmond	.....	WP	DB	OS
Shannon	Black	4141 Lake City Way	Seattle	.....	DB	SS	
Estela	Rosales	970 Phoenix Avenue	Bellevue	.....	DTP	WP	PG

Figure 7.18 *The result of "flattening" the* CATEGORIES TAUGHT *field*

Commas separate the occurrences within this value.

Flattening is an Ok solution?

Instructors

InstFirst Name	InstLast Name	InstStreet Address	InstCity	<< other fields >>	Categories Taught
Kira	Bently	3131 Mockingbird Lane	Seattle	.....	DTP, SS, WP
Timothy	Ennis	7402 Kingman Drive	Redmond	.....	WP, DB, OS
Shannon	Black	4141 Lake City Way	Seattle	.....	DB, SS
Estela	Rosales	970 Phoenix Avenue	Bellevue	.....	DTP, WP, PG

Figure 7.17 *Identifying a multivalued field*

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Resolve multivalued fields (sol. 2)

Instructors

InstFirst Name	InstLast Name	InstStreet Address	InstCity	InstState	InstZipcode	InstPhone	Categories Taught
Kira	Bently	3131 Mockingbird Lane	Seattle	WA	98157	363-9948	DTP
Kira	Bently	3131 Mockingbird Lane	Seattle	WA	98157	363-9948	SS
Kira	Bently	3131 Mockingbird Lane	Seattle	WA	98157	363-9948	WP
Timothy	Ennis	7402 Kingman Drive	Redmond	WA	98115	527-4992	WP
Timothy	Ennis	7402 Kingman Drive	Redmond	WA	98115	527-4992	DB
Timothy	Ennis	7402 Kingman Drive	Redmond	WA	98115	527-4992	OS
Shannon	Black	4141 Lake City Way	Seattle	WA	98136	336-5992	DB
Shannon	Black	4141 Lake City Way	Seattle	WA	98136	336-5992	SS
Estela	Rosales	970 Phoenix Avenus	Bellevue	WA	98046	322-6992	DTP
Estela	Rosales	970 Phoenix Avenus	Bellevue	WA	98046	322-6992	WP
Estela	Rosales	970 Phoenix Avenus	Bellevue	WA	98046	322-6992	PG

Any faults with this solution?

Figure 7.19 *The result of bringing* CATEGORIES TAUGHT *into compliance with the second element of an ideal field*

Commas separate the occurrences within this value.

Instructors

InstFirst Name	InstLast Name	InstStreet Address	InstCity	*<< other fields >>*	Categories Taught
Kira	Bently	3131 Mockingbird Lane	Seattle	.....	DTP, SS, WP
Timothy	Ennis	7402 Kingman Drive	Redmond	.....	WP, DB, OS
Shannon	Black	4141 Lake City Way	Seattle	.....	DB, SS
Estela	Rosales	970 Phoenix Avenue	Bellevue	.....	DTP, WP, PG

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Resolve multivalued fields (sol. 3)

Instructors

InstFirst Name	InstLast Name	InstStreet Address	InstCity	InstState	InstZipcode	InstPhone
Kira	Bently	3131 Mockingbird Lane	Seattle	WA	98157	363-9948
Timothy	Ennis	7402 Kingman Drive	Redmond	WA	98115	527-4992
Shannon	Black	4141 Lake City Way	Seattle	WA	98136	336-5992
Estela	Rosales	970 Phoenix Avenue	Bellevue	WA	98046	322-6992

Instructor Categories

InstFirst Name	InstLast Name	Category Taught
Kira	Bently	DTP
Kira	Bently	SS
Kira	Bently	WP
Timothy	Ennis	WP
Timothy	Ennis	DB
Timothy	Ennis	OS
Shannon	Black	DB
Shannon	Black	SS

Commas separate the occurrences within this value.

Instructors

InstFirst Name	InstLast Name	InstStreet Address	InstCity	*<< other fields >>*	Categories Taught
Kira	Bently	3131 Mockingbird Lane	Seattle	.....	DTP, SS, WP
Timothy	Ennis	7402 Kingman Drive	Redmond	.....	WP, DB, OS
Shannon	Black	4141 Lake City Way	Seattle	.....	DB, SS
Estela	Rosales	970 Phoenix Avenue	Bellevue	.....	DTP, WP, PG

Figure 7.20 *Resolving the multivalued field in the INSTRUCTORS table*

Step 1) Field becomes **new table**

Step 2) Use a set of fields to **connect the two tables** (use minimum redundant data)

Step 3) Create **name, type, description** for the new table

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Another ex. Resolve multivalued fields

Instructors

InstFirst Name	InstLast Name	Campus Phone	Categories Taught	Maximum Level Taught	Languages Spoken
Kira	Bently	363-9948	DTP, OS, SS, WP	Intermediate, Basic, Advanced, Basic	French, Spanish
Timothy	Ennis	527-4992	DB, OS, UT, WP	Intermediate, Basic, Basic, Advanced	German, Spanish
Shannon	Black	336-5992	DB, PG, SS	Advanced, Intermediate, Intermediate	French, German
Estela	Rosales	322-6992	DTP, PG, WP	Basic, Intermediate, Basic	French, Italian, Spanish

Figure 7.21 *A version of the INSTRUCTORS table containing three multivalued fields*

Instructors

InstFirst Name	InstLast Name	Campus Phone
Kira	Bently	363-9948
Timothy	Ennis	527-4992
Shannon	Black	336-5992
Estela	Rosales	322-6992

Instructor Categories

InstFirst Name	InstLast Name	Category Taught	Maximum Level
Kira	Bently	DTP	Intermediate
Kira	Bently	OS	Advanced
Kira	Bently	SS	Basic
Kira	Bently	WP	Advanced
Timothy	Ennis	DB	Intermediate
Timothy	Ennis	OS	Basic
Timothy	Ennis	UT	Basic
Timothy	Ennis	WP	Advanced

Instructor Languages

InstFirst Name	InstLast Name	Language Spoken
Kira	Bently	French
Kira	Bently	Spanish
Timothy	Ennis	German
Timothy	Ennis	Spanish
Shannon	Black	French
Shannon	Black	German
Estela	Rosales	French
Estela	Rosales	Italian
Estela	Rosales	Spanish

Figure 7.22 *Resolving the multivalued fields in the INSTRUCTORS table*

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Part E) Elements of the ideal table

1. Represents a single subject.
2. Has a primary key
3. Has no multipart or multivalued fields
4. Has no calculated fields
5. Has no duplicate fields
6. Contains the minimum amount of redundant data

Remember elements of the ideal field:

1. Represents distinct characteristic of the subject
2. Contains only a single value – ex. not multivalued “phone/fax”
3. Cannot be deconstructed into smaller parts – ex. not multipart “violin, piano”
4. Is not a calculated field
5. Distinct in the entire database
6. Retains its properties when appears in other tables

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Ex1: Resolve duplicate fields

These fields duplicate the MANPHONE and WEB SITE fields in the MANUFACTURERS table.

Instruments

Instrument ID	Instrument Description	Category	Price	Manufacturer	ManPhone	Web Site
2201	Stratocaster	Guitar	$ 799.99	Fender Musical Instruments	596-9690	www.fender.com
3349	Player 2100 Multi-Effects	Multi-Effect Unit	$ 174.99	Samson Technologies Corp.	364-2244	www.samsontech.com
1001	JCM 2000 Tube Super Lead	Amplifier	$ 549.99	Mesa/Boogie	778-6565	www.mesaboogie.com
5590	Crate VC60 Pro Tube Amp	Amplifier	$ 399.99	St. Louis Music, Inc.	738-7563	www.crateamps.com
2227	Cry Baby Wah-Wah	Single-Effect Unit	$ 169.99	Dunlop Manufacturing, Inc.	745-2722	www.jimdunlop.com
2766	Twin Reverb Reissue	Amplifier	$ 1,224.99	Fender Musical Instruments	596-9690	www.fender.com

Manufacturers

Manufacturer	ManStreet Address	ManCity	ManState	ManZipcode	ManPhone	Web Site
Dunlop Manufacturing, Inc.	PO Box 846	Benicia	CA	94510	745-2722	www.jimdunlop.com
Fender Musical Instruments	8860 E. Chaparral Road	Scottsdale	AZ	85250	596-9690	www.fender.com
Mesa/Boogie	1317 Ross Street	Petaluma	CA	94954	778-6565	www.mesaboogie.com
Samson Technologies Corp.	PO Box 9031	Syosset	NY	11791	364-2244	www.samsontech.com
St. Louis Music, Inc.	1400 Ferguson Avenue	St. Louis	MO	63133	738-7563	www.crateamps.com

Figure 7.23 *Example of a table containing reference fields*

Sol.: Eliminate redundant data, then combine information from the two tables in a view to create reports

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Ex2: Resolve duplicate fields

Figure 7.24 *A simple example of a table containing unnecessary duplicate fields*

Students

StdFirst Name	StdLast Name	StdStreet Address	<< other fields >>	Instrument 1	Instrument 2	Instrument 3
Scott	Baker	2904 Madison Ave	.....	Guitar	Tenor Sax	
Michael	Chow	7410 Taxco Drive	.....	Tenor Sax	Clarinet	Electric Piano
Debbie	McGuire	332 158th Ave SE	.....	Drum Set	Bass Guitar	
Angie	Thomson	970 Pine Blvd	.....	Guitar	Electric Piano	Snare Drum

These duplicate fields represent three occurrences of the same *type* of value.

Students

StdFirst Name	*StdLast Name*	*StdStreet Address*	*<< other fields >>*	***Instruments***
Scott	*Baker*	*2904 Madison Ave*	.....	***Guitar, Tenor Sax***
Michael	*Chow*	*7410 Taxco Drive*	.....	***Tenor Sax, Clarinet, Electric Piano***
Debbie	*McGuire*	*332 158th Ave SE*	.....	***Drum Set, Bass Guitar***
Angie	*Thomson*	*970 Pine Blvd*	.....	***Guitar, Electric Piano, Snare Drum***

Students

StdFirst Name	StdLast Name	StdStreet Address	<< other fields >>
Scott	Baker	2904 Madison Ave	.....
Michael	Chow	7410 Taxco Drive	.....
Debbie	McGuire	332 158th Ave SE	.....
Angie	Thomson	970 Pine Blvd	.....

Student Instruments

StudFirst Name	StudLast Name	Instrument
Scott	Baker	Guitar
Scott	Baker	Tenor Sax
Michael	Chow	Tenor Sax
Michael	Chow	Clarinet
Michael	Chow	Electric Piano
Debbie	McGuire	Drum Set
Debbie	McGuire	Bass Guitar

Advantages:

1) Student can play any number of instruments

2) Easy to retrieve information (ex. All students playing Piano)

Figure 7.25 *Resolving a simple set of unnecessary duplicate fields*

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Ex3: Resolve duplicate fields

Students

StdFirst Name	StdLast Name	*<< other fields >>*	Instrument 1	Checkout Date 1	Instrument 2	Checkout Date 2	Instrument 3	Checkout Date 3
Scott	Baker	.....	Guitar	09/26/01	Tenor Sax	09/28/01		
Michael	Chow	.....	Tenor Sax	09/26/01	Clarinet	10/03/01	Electric Piano	10/16/12
Debbie	McGuire	.....	Drum Set	11/14/01	Bass Guitar	11/20/01		
Angie	Thomson	.....	Guitar	11/14/01	Electric Piano	11/14/01	Snare Drum	12/05/12

Figure 7.26 *Example of a table with multiple sets of duplicate fields*

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Students

StdFirst Name	StdLast Name	*<< other fields >>*	Instrument 1	Checkout Date 1	Instrument 2	Checkout Date 2	Instrument 3	Checkout Date 3
Scott	Baker	.....	Guitar	09/26/01	Tenor Sax	09/28/01		
Michael	Chow	.....	Tenor Sax	09/26/01	Clarinet	10/03/01	Electric Piano	10/16/12
Debbie	McGuire	.....	Drum Set	11/14/01	Bass Guitar	11/20/01		
Angie	Thomson	.....	Guitar	11/14/01	Electric Piano	11/14/01	Snare Drum	12/05/12

Students

StdFirst Name	*StdLast Name*	*<< other fields >>*	***Instruments***	***Checkout Dates***
Scott	*Baker*	.....	***Guitar, Tenor Sax***	***09/26/12, 09/28/12***
Michael	*Chow*	.....	***Tenor Sax, Clarinet, Electric Piano***	***09/28/12, 10/03/12, 10/16/12***
Debbie	*McGuire*	.....	***Drum Set, Bass Guitar***	***11/14/12, 11/20/12***
Angie	*Thomson*	.....	***Guitar, Electric Piano, Snare Durm***	***11/14/12, 11/14/12, 12/05/12***

Students

StdFirst Name	StdLast Name	StdStreet Address	*<< other fields >>*
Scott	Baker	2904 Madison Ave	.....
Michael	Chow	7410 Taxco Drive	.....
Debbie	McGuire	332 158th Ave SE	.....
Angie	Thomson	970 Pine Blvd	.....

Student Instruments

StudFirst Name	StudLast Name	Instrument	Checkout Date
Scott	Baker	Guitar	09/26/12
Scott	Baker	Tenor Sax	09/28/12
Michael	Chow	Tenor Sax	09/28/12
Michael	Chow	Clarinet	10/03/12
Michael	Chow	Electric Piano	10/16/12
Debbie	McGuire	Drum Set	11/14/12
Debbie	McGuire	Bass Guitar	11/20/12

Figure 7.27 *Resolving the multiple sets of duplicate fields in the STUDENTS table*

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Part F) Create subset tables

- **Problem:** if having too many blanks in a table ➔ table represents 2+ subjects
- **Solution:** create subset tables

Table Structures

Inventory	
Item Name	Model
Item Description	Warranty Expiration Date
Current Value	Publisher
Insured Value	Author
Date Entered	ISBN
Manufacturer	Category

Figure 7.28 *Structure of an office inventory table*

Table Structures

Inventory	Equipment	Books
Item Name	Item Name	Item Name
Item Description	Manufacturer	Publisher
Current Value	Model	Author
Insured Value	Warranty Expiration Date	ISBN
Date Entered		Category

Figure 7.29 *The new subset table structures*

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Another example of subset tables

Table Structures

Full-Time Employees

FTEFirst Name
FTELast Name
FTEDate Hired
Salary Amount
Position
FTEStreet Address
FTECity
FTEState

Table Structures

Part-Time Employees

PTEFirst Name
PTELast Name
PTEDate Hired
Hourly Rate
Skill Level
PTEStreet Address
PTECity
PTEState

Figure 7.30 *Previously unidentified subset tables*

Table Structures

Employees	Full-Time Employees	Part-Time Employees
EmpFirst Name	EmpFirst Name	EmpFirst Name
EmpLast Name	EmpLast Name	EmpLast Name
Date Hired	Salary Amount	Hourly Rate
EmpStreet Address	Position	Skill Level
EmpCity		
EmpState		

Figure 7.31 *The results of refining the subset tables*

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Covered in this chapter:

- Part A) Three steps to **create preliminary table list**
- Part B) Create **name, type, description** for tables
- Part C) **Associate fields** with each table
- Part D) Elements of the **ideal field**
- Part E) Elements of the **ideal table**
- Part F) Create **subset tables**

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